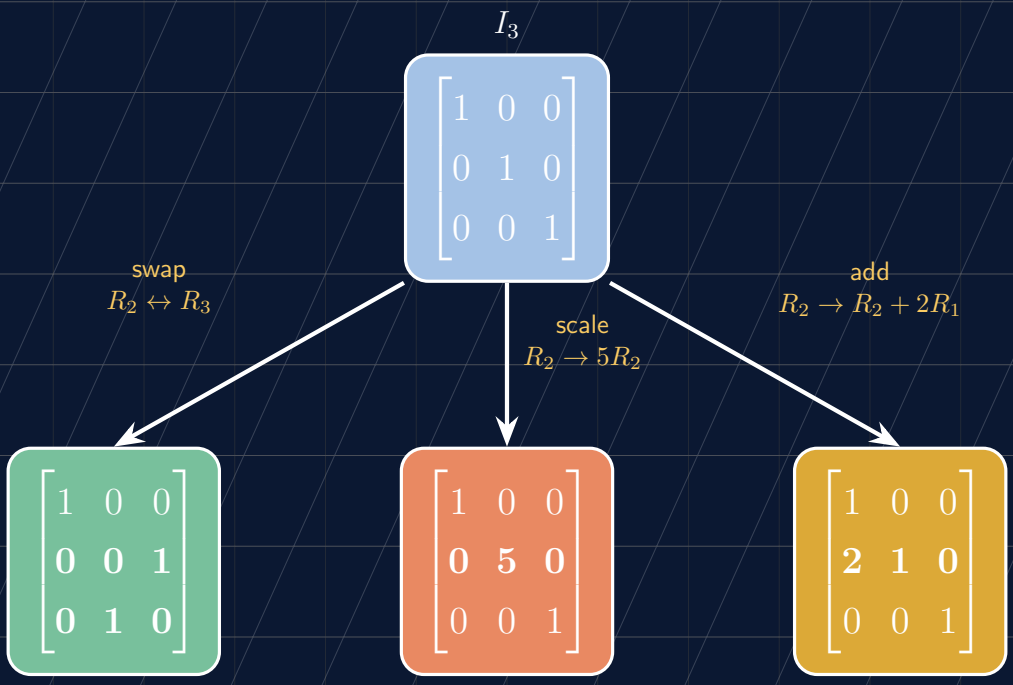
LINEAR ALGEBRA · LECTURE 5

Elementary Matrices

and a method for finding A^{-1}

Row operations as matrices · Inverses · The inversion algorithm
LU decomposition · Cryptography · Colour models · Geometry in 2D and 3D



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Lecture notes for BS students | every row operation written out, many pictures, a summary sheet

Built on: Anton & Rorres, *Elementary Linear Algebra* (§1.5) · G. Williams, *Linear Algebra with Applications* (§2.4)

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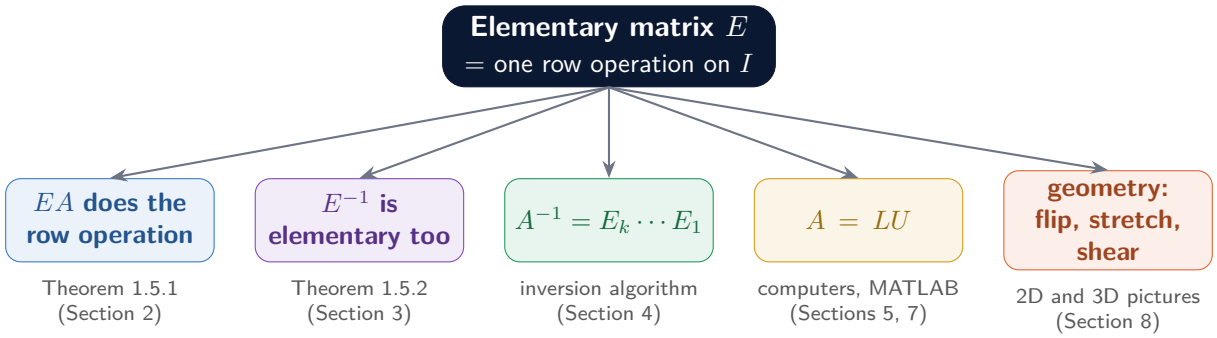
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How to read these notes

- Every row operation is written **above** the arrow, and the name of its elementary matrix **below** it: $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \xrightarrow[E_1]{R_2 \to R_2 - 3R_1} \begin{bmatrix} 1 & 2 \\ \mathbf{0} & \mathbf{-2} \end{bmatrix}$.
- The row that has just **changed** is printed in **orange**. Nothing is skipped: every intermediate matrix is shown.
- Notation: R_i = row i. $R_i \leftrightarrow R_j$ = swap rows i and j. $R_i \to cR_i$ = multiply row i by $c \neq 0$. $R_i \to R_i + cR_j$ = add c times row j to row i.
- Boxes: Definition | In simple words | Theorem | Example | Common mistake | Where is this used?
- Tags such as [Anton §1.5 · Ex. 4] tell you where the example comes from in the textbook.



1 What is an elementary matrix?

1.1 The three elementary row operations

In Lecture 1 we solved linear systems with three simple moves on the rows of a matrix.

Type	Operation	Symbol	Example on $\begin{bmatrix}1 & 2\\3 & 4\end{bmatrix}$
1. Swap	interchange two rows	$R_i \leftrightarrow R_j$	$\begin{bmatrix}1 & 2\\3 & 4\end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix}\mathbf{3} & \mathbf{4}\\\mathbf{1} & \mathbf{2}\end{bmatrix}$
2. Scale	multiply a row by a constant $c \neq 0$	$R_i \to cR_i$	$\begin{bmatrix}1 & 2\\3 & 4\end{bmatrix} \xrightarrow{R_2 \to 5R_2} \begin{bmatrix}1 & 2\\\mathbf{15} & \mathbf{20}\end{bmatrix}$
3. Add	add c times one row to another row	$R_i \to R_i + cR_j$	$\begin{bmatrix}1 & 2\\3 & 4\end{bmatrix} \xrightarrow{R_2 \to R_2 - 3R_1} \begin{bmatrix}1 & 2\\\mathbf{0} & \mathbf{-2}\end{bmatrix}$

Each move can be **undone** by a move of the same type (multiply by $1/c$; swap again; subtract instead of add). This simple fact is behind everything in this lecture.

Definition 1 (Row equivalent) [Anton §1.5 · Def. 1]

Matrices A and B are **row equivalent** if either (hence each) can be obtained from the other by a sequence of elementary row operations.

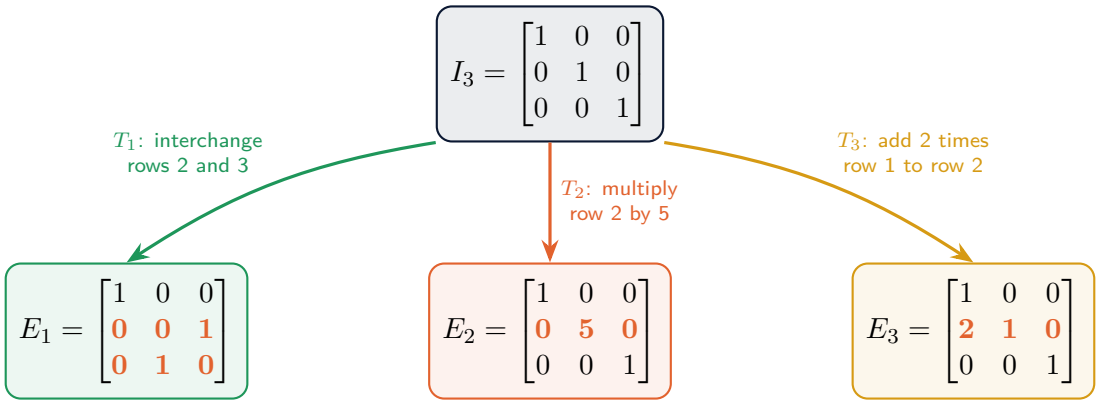
1.2 The definition

Definition 2 (Elementary matrix) [Anton §1.5 · Def. 2]

A matrix E is called an **elementary matrix** if it can be obtained from an identity matrix by performing a **single** elementary row operation.

In simple words

Take the identity matrix I. Do *exactly one* row operation to it. Whatever you get is an elementary matrix. Since there are three kinds of row operation, there are three kinds of elementary matrix: a **swap** matrix, a **scale** matrix and an **add** matrix. Elementary matrices are always **square**.



one move from I $\Longrightarrow$ one elementary matrix [Williams §2.4 · Illustration]

Example 1 Elementary matrices and row operations [Anton §1.5 · Ex. 1]

Listed below are four elementary matrices and the single operation that produces each one.

$$\underbrace{\begin{bmatrix}1&0\\0&-3\end{bmatrix}}_{\text{multiply the second row of } I_2 \text{ by } -3} \qquad \underbrace{\begin{bmatrix}1&0&0&0\\0&0&0&1\\0&0&1&0\\0&1&0&0\end{bmatrix}}_{\text{interchange the second and fourth rows of } I_4} \qquad \underbrace{\begin{bmatrix}1&0&3\\0&1&0\\0&0&1\end{bmatrix}}_{\text{add 3 times the third row of } I_3 \text{ to the first}} \qquad \underbrace{\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}}_{\text{multiply the first row of } I_3 \text{ by 1}}$$

Step by step for the third one:

$$\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix} \xrightarrow{R_1 \to R_1 + 3R_3} \begin{bmatrix}\mathbf{1}&\mathbf{0}&\mathbf{3}\\0&1&0\\0&0&1\end{bmatrix}, \qquad \text{because } R_1 + 3R_3 = (1,0,0) + 3(0,0,1) = (1,0,3).$$

Note the last one: I itself is elementary (multiply a row by $c = 1$; or add 0 times a row).

1.3 How to recognise an elementary matrix

The three-question test

Compare the square matrix E with the identity matrix of the same size.

1. **Swap?** Two rows of I are interchanged and *nothing else* changed.
2. **Scale?** Exactly one diagonal 1 has been replaced by a number $c \neq 0$, nothing else changed.
3. **Add?** Exactly one off-diagonal 0 has been replaced by a number c, nothing else changed.

If none of the three describes E, then E is **not** elementary. Two changes = two operations = not elementary.

swap

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

two rows of I exchanged

scale

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & c & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

one diagonal entry $c \neq 0$

add

$$\begin{bmatrix} 1 & 0 & 0 \\ c & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

one off-diagonal entry c

Example **Is it elementary?** [Anton §1.5 · Ex. Set 1.5, No. 1]

	Matrix	Verdict	Reason (compare with I)
(a)	$\begin{bmatrix} 1 & 0 \\ -5 & 1 \end{bmatrix}$	yes	One off-diagonal 0 became -5: $I_2 \xrightarrow{R_2 \to R_2 - 5R_1}$ this matrix.
(b)	$\begin{bmatrix} -5 & 1 \\ 1 & 0 \end{bmatrix}$	no	Row 1 is $(-5, 1)$: no single move on I_2 makes this. (A swap gives $(0, 1)$; then we would still need $R_1 \to R_1 - 5R_2$: two moves.)
(c)	$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$	no	It has a **zero row**. Scaling needs $c \neq 0$, so no single operation on I_3 produces a zero row.
(d)	$\begin{bmatrix} 2 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	no	Row 1 of I_4 changed in **two** places: the diagonal $1 \to 2$ *and* a $0 \to 2$. That needs two moves ($R_1 \to 2R_1$, then $R_1 \to R_1 + 2R_4$).

Example **Is it elementary?** [Anton §1.5 · Ex. Set 1.5, No. 2]

	Matrix	Verdict	Reason (compare with I)
(a)	$\begin{bmatrix} 1 & 0 \\ 0 & \sqrt{3} \end{bmatrix}$	yes	Scale: $I_2 \xrightarrow{R_2 \to \sqrt{3}\,R_2}$ this matrix ($\sqrt{3} \neq 0$).
(b)	$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$	yes	Swap: $I_3 \xrightarrow{R_1 \leftrightarrow R_3}$ this matrix.
(c)	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 9 \\ 0 & 0 & 1 \end{bmatrix}$	yes	Add: $I_3 \xrightarrow{R_2 \to R_2 + 9R_3}$ this matrix, since $(0, 1, 0) + 9(0, 0, 1) = (0, 1, 9)$.
(d)	$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$	no	Two moves are needed: $R_1 \to -R_1$ *and* $R_2 \leftrightarrow R_3$.

Common mistake

- A zero row can never appear in an elementary matrix (scaling by $c = 0$ is *not* allowed).
- "Looks simple" is not enough: $\begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix}$ is *not* elementary (it is a swap *and* a scale).
- The product of two elementary matrices is usually *not* elementary [Anton TF (a)].

2 Why does EA perform the row operation?

Theorem 1.5.1 Row operations by matrix multiplication [Anton §1.5]

If the elementary matrix E results from performing a certain row operation on I_m and if A is an $m \times n$ matrix, then the product EA is the matrix that results when this **same row operation** is performed on A.

In simple words

To do a row operation on A you may either (i) do it directly, or (ii) do it to I first (this gives E), and then multiply E on the **left** of A. Both roads give the same answer:

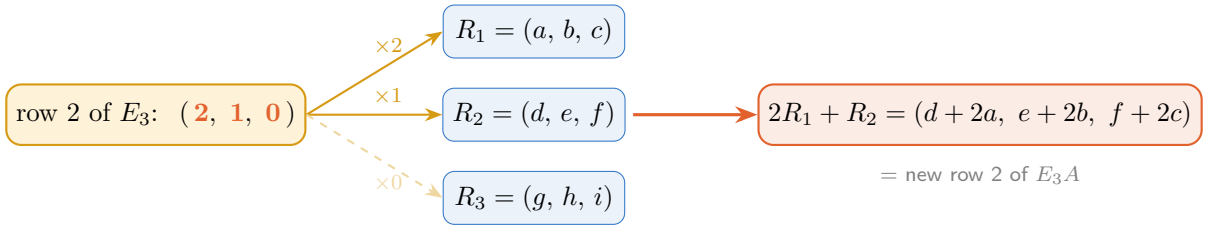
$$\text{row operation on } A \;=\; E\,A, \qquad \text{where } E = (\text{same row operation on } I).$$

2.1 The reason: each row of E is a recipe

Recall how a product is built: **row** i **of** $EA =$ (row i of E) $\times A$, and

$$(e_1,\, e_2,\, e_3)\begin{bmatrix} — & R_1 & — \\ — & R_2 & — \\ — & R_3 & — \end{bmatrix} \;=\; e_1R_1 + e_2R_2 + e_3R_3.$$

So row i of E is a *recipe*: it says how much of each row of A to mix into row i of the answer. A row of I, for example $(0, 1, 0)$, is the recipe "take row 2 and nothing else". That is why the rows of E that were not touched leave the rows of A untouched.



2.2 All three types on a general 3×3 matrix

[Williams §2.4 · Illustration]

Let $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$ and use the three elementary matrices E_1, E_2, E_3 of Section 1.

Type 1 — interchange rows 2 and 3. The rows of E_1 are the recipes $(1,0,0)$, $(0,0,1)$, $(0,1,0)$:

$$E_1 A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} 1a+0d+0g & 1b+0e+0h & 1c+0f+0i \\ 0a+0d+1g & 0b+0e+1h & 0c+0f+1i \\ 0a+1d+0g & 0b+1e+0h & 0c+1f+0i \end{bmatrix}$$

$$= \begin{bmatrix} a & b & c \\ \mathbf{g} & \mathbf{h} & \mathbf{i} \\ \mathbf{d} & \mathbf{e} & \mathbf{f} \end{bmatrix}.$$

Type 2 — multiply row 2 by 5.

$$E_2 A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} 1a+0d+0g & 1b+0e+0h & 1c+0f+0i \\ 0a+5d+0g & 0b+5e+0h & 0c+5f+0i \\ 0a+0d+1g & 0b+0e+1h & 0c+0f+1i \end{bmatrix}$$

$$= \begin{bmatrix} a & b & c \\ \mathbf{5d} & \mathbf{5e} & \mathbf{5f} \\ g & h & i \end{bmatrix}.$$

Type 3 — add 2 times row 1 to row 2.

$$E_3 A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} 1a+0d+0g & 1b+0e+0h & 1c+0f+0i \\ 2a+1d+0g & 2b+1e+0h & 2c+1f+0i \\ 0a+0d+1g & 0b+0e+1h & 0c+0f+1i \end{bmatrix}$$

$$= \begin{bmatrix} a & b & c \\ \mathbf{d+2a} & \mathbf{e+2b} & \mathbf{f+2c} \\ g & h & i \end{bmatrix}.$$

In each case the answer is exactly A with the row operation done to it. Since A was *any* matrix, this is really a proof of Theorem 1.5.1 for 3×3 matrices; the general proof is the same idea.

Example 2 Using elementary matrices [Anton §1.5 · Ex. 2]

Consider

$$A = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 2 & -1 & 3 & 6 \\ 1 & 4 & 4 & 0 \end{bmatrix}, \qquad E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix} \quad (\text{from } I_3 \xrightarrow{R_3 \to R_3 + 3R_1}).$$

Rows 1 and 2 of EA. Rows 1 and 2 of E are $(1,0,0)$ and $(0,1,0)$, the recipes "copy row 1" and "copy row 2". So rows 1 and 2 of EA are $(1,0,2,3)$ and $(2,-1,3,6)$.
Row 3 of EA. Row 3 of E is $(3,0,1)$. Multiply it by each column of A:

column 1: $3(1)+0(2)+1(1) = 3+0+1 = 4$, column 2: $3(0)+0(-1)+1(4) = 4$,
column 3: $3(2)+0(3)+1(4) = 6+0+4 = 10$, column 4: $3(3)+0(6)+1(0) = 9$.

$$EA = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 2 & -1 & 3 & 6 \\ \mathbf{4} & \mathbf{4} & \mathbf{10} & \mathbf{9} \end{bmatrix}.$$

Check directly: $R_3 + 3R_1 = (1,4,4,0) + 3(1,0,2,3) = (1+3,\ 4+0,\ 4+6,\ 0+9) = (4,4,10,9)$.
✓ So EA is precisely the matrix obtained by adding 3 times the first row of A to the third row.

Common mistake

Left, not right! EA does a *row* operation. AE does a *column* operation. Example with $E = \begin{bmatrix}1 & 0\\ 3 & 1\end{bmatrix}$ (from $R_2 \to R_2 + 3R_1$) and $A = \begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}$:

$$EA = \begin{bmatrix}1 & 0\\ 3 & 1\end{bmatrix}\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix} = \begin{bmatrix}1 & 2\\ \mathbf{6} & \mathbf{10}\end{bmatrix} \qquad \text{(row } 2 + 3 \times \text{row } 1\text{)},$$

$$AE = \begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}\begin{bmatrix}1 & 0\\ 3 & 1\end{bmatrix} = \begin{bmatrix}\mathbf{7} & 2\\ \mathbf{15} & 4\end{bmatrix} \qquad \text{(column } 1 + 3 \times \text{column } 2\text{)}.$$

Also: if A is $m \times n$, then E must be $m \times m$ (it acts on the m rows).

Where is this used?

Theorem 1.5.1 turns "doing things to rows" into "multiplying matrices". That is what makes it possible to *prove* results about row reduction with algebra (Sections 4 and 7), and it is how computers store a whole elimination as a single matrix (the LU decomposition, Section 5.4). In hand calculations it is still quicker to perform row operations directly [Anton, margin note].

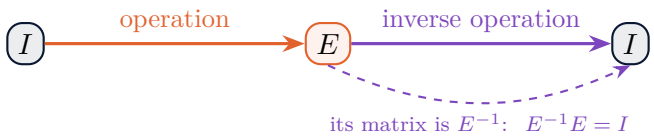
3 The inverse of an elementary matrix

3.1 Every move can be undone

If B is obtained from A by one row operation, then A can be recovered from B by the **inverse operation** [Anton §1.5]:

Operation on I that produces E	Operation on E that gives back I	Rule
Multiply row i by $c \neq 0$	Multiply row i by $1/c$	**scale → reciprocal**
Interchange rows i and j	Interchange rows i and j	**swap → itself**
Add c times row i to row j	Add $-c$ times row i to row j	**add → change sign**

Table 1 [Anton §1.5 · Table 1]



Example 3 Row operations and inverse row operations [Anton §1.5 · Ex. 3]

In each line an operation is applied to I_2 to obtain an elementary matrix E, then E is restored to I_2 by the inverse operation.

$$\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} \xrightarrow{R_2 \to 7R_2} \begin{bmatrix}1 & 0\\ \mathbf{0} & \mathbf{7}\end{bmatrix} \xrightarrow{R_2 \to \frac{1}{7}R_2} \begin{bmatrix}1 & 0\\ \mathbf{0} & \mathbf{1}\end{bmatrix} \qquad \text{scale by 7, then by } \tfrac{1}{7}$$

$$\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix}\mathbf{0} & \mathbf{1}\\ \mathbf{1} & \mathbf{0}\end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix}\mathbf{1} & \mathbf{0}\\ \mathbf{0} & \mathbf{1}\end{bmatrix} \qquad \text{swap, then swap again}$$

$$\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} \xrightarrow{R_1 \to R_1 + 5R_2} \begin{bmatrix}\mathbf{1} & \mathbf{5}\\ 0 & 1\end{bmatrix} \xrightarrow{R_1 \to R_1 - 5R_2} \begin{bmatrix}\mathbf{1} & \mathbf{0}\\ 0 & 1\end{bmatrix} \qquad \text{add } 5R_2\text{, then add } -5R_2$$

So, for instance, $\begin{bmatrix}1 & 5\\ 0 & 1\end{bmatrix}^{-1} = \begin{bmatrix}1 & -5\\ 0 & 1\end{bmatrix}$. Check:

$$\begin{bmatrix}1 & 5\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & -5\\ 0 & 1\end{bmatrix} = \begin{bmatrix}1\cdot 1+5\cdot 0 & 1(-5)+5\cdot 1\\ 0\cdot 1+1\cdot 0 & 0(-5)+1\cdot 1\end{bmatrix} = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}. \checkmark$$

Theorem 1.5.2 [Anton §1.5]

Every elementary matrix is invertible, and the inverse is also an elementary matrix.

Proof. If E is an elementary matrix, then E results from performing some row operation on I. Let E_0 be the matrix that results when the *inverse* of that operation is performed on I. By Theorem 1.5.1, E_0E is "do the inverse operation to E", which gives back I; and EE_0 is "do the operation to E_0", which also gives I, because inverse operations cancel each other:

$$E_0E = I \qquad \text{and} \qquad EE_0 = I.$$

Thus the elementary matrix E_0 is the inverse of E. ■

Write E^{-1} in one second

$$\underbrace{\begin{bmatrix}1 & 0 & 0\\ 0 & c & 0\\ 0 & 0 & 1\end{bmatrix}^{-1} = \begin{bmatrix}1 & 0 & 0\\ 0 & \frac{1}{c} & 0\\ 0 & 0 & 1\end{bmatrix}}_{\text{scale: take the reciprocal}} \qquad \underbrace{\begin{bmatrix}1 & 0 & 0\\ k & 1 & 0\\ 0 & 0 & 1\end{bmatrix}^{-1} = \begin{bmatrix}1 & 0 & 0\\ -k & 1 & 0\\ 0 & 0 & 1\end{bmatrix}}_{\text{add: change the sign}} \qquad \underbrace{\begin{bmatrix}0 & 1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 1\end{bmatrix}^{-1} = \begin{bmatrix}0 & 1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 1\end{bmatrix}}_{\text{swap: its own inverse}}$$

Example Quick inverses [Williams §2.4 · Ex. Set 2.4, No. 33]

For each matrix: name the row operation, name its inverse operation, write the inverse matrix.

	E	operation on I_3	inverse operation	E^{-1}
(a)	$\begin{bmatrix}0 & 0 & 1\\ 0 & 1 & 0\\ 1 & 0 & 0\end{bmatrix}$	$R_1 \leftrightarrow R_3$	$R_1 \leftrightarrow R_3$	$\begin{bmatrix}0 & 0 & 1\\ 0 & 1 & 0\\ 1 & 0 & 0\end{bmatrix}$
(b)	$\begin{bmatrix}0 & 1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 1\end{bmatrix}$	$R_1 \leftrightarrow R_2$	$R_1 \leftrightarrow R_2$	$\begin{bmatrix}0 & 1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 1\end{bmatrix}$
(c)	$\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 4\end{bmatrix}$	$R_3 \to 4R_3$	$R_3 \to \frac{1}{4}R_3$	$\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & \frac{1}{4}\end{bmatrix}$
(d)	$\begin{bmatrix}\frac{1}{3} & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$	$R_1 \to \frac{1}{3}R_1$	$R_1 \to 3R_1$	$\begin{bmatrix}3 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$
(e)	$\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ -4 & 0 & 1\end{bmatrix}$	$R_3 \to R_3 - 4R_1$	$R_3 \to R_3 + 4R_1$	$\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 4 & 0 & 1\end{bmatrix}$
(f)	$\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 3 & 1\end{bmatrix}$	$R_3 \to R_3 + 3R_2$	$R_3 \to R_3 - 3R_2$	$\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & -3 & 1\end{bmatrix}$

Rules (Williams): a swap matrix is its own inverse; for a scale matrix replace c by $1/c$; for an add matrix change the sign of the off-diagonal entry. Check (e): in the product below, rows 1 and 2 are copied, and row 3 is $(-4)(1,0,0)+0(0,1,0)+1(4,0,1)=(0,0,1)$:

$$\begin{bmatrix}1&0&0\\0&1&0\\-4&0&1\end{bmatrix}\begin{bmatrix}1&0&0\\0&1&0\\4&0&1\end{bmatrix}=\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}=I_3.\ \checkmark$$

Common mistake

The inverse of "add c times row i to row j" is *not* "subtract row j from row i". Keep the **same two rows in the same roles** and only change the sign of c.

4 Elementary matrices *from* the inverse: the inversion algorithm

4.1 The equivalence theorem

Theorem 1.5.3 Equivalent statements [Anton §1.5]

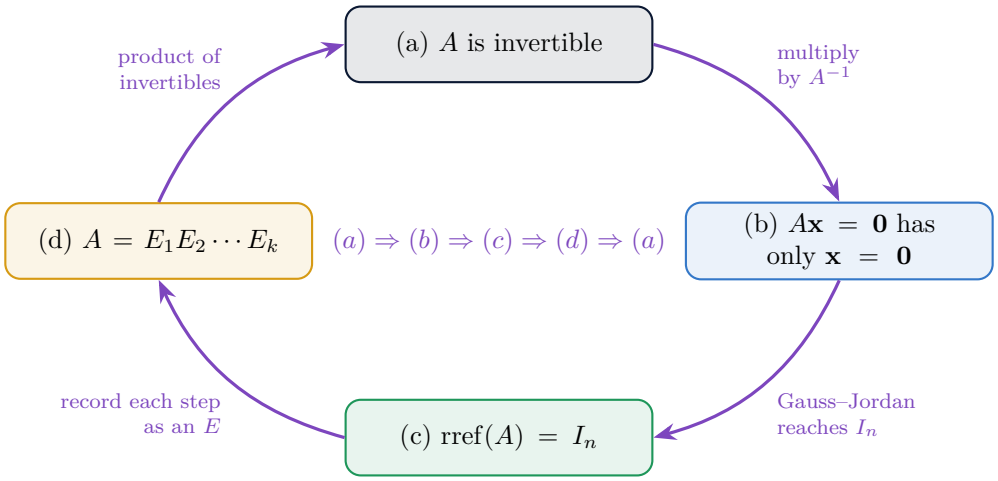
If A is an $n\times n$ matrix, then the following statements are equivalent, that is, all true or all false.

(a) A is invertible.

(b) $A\mathbf{x}=\mathbf{0}$ has only the trivial solution.

(c) The reduced row echelon form of A is I_n.

(d) A is expressible as a product of elementary matrices.



A closed cycle of implications means every statement implies every other one, so they are all equivalent.

Proof.

(a)⇒(b) If A is invertible and $A\mathbf{x}_0=\mathbf{0}$, multiply by A^{-1}: $A^{-1}A\mathbf{x}_0=A^{-1}\mathbf{0}$, so $\mathbf{x}_0=\mathbf{0}$.

(b)⇒(c) If $A\mathbf{x}=\mathbf{0}$ has only the trivial solution, Gauss–Jordan elimination on $[\,A\mid\mathbf{0}\,]$ must end with $x_1=0,\ x_2=0,\ \ldots,\ x_n=0$, i.e. with $[\,I_n\mid\mathbf{0}\,]$. Ignoring the last column (all zeros), $\mathrm{rref}(A)=I_n$.

(c)⇒(d) If A reduces to I_n by row operations, then by Theorem 1.5.1 each operation is a multiplication on the left by an elementary matrix, so there are $E_1, \dots, E_k$ with

$$E_k \cdots E_2 E_1 A = I_n. \tag{3}$$

By Theorem 1.5.2 each E_i is invertible. Multiply (3) on the left by E_k^{-1}, then E_{k-1}^{-1}, …, then E_1^{-1}:

$$A = E_1^{-1} E_2^{-1} \cdots E_k^{-1} I_n = E_1^{-1} E_2^{-1} \cdots E_k^{-1}, \tag{4}$$

a product of elementary matrices (Theorem 1.5.2 again).

(d)⇒(a) A product of invertible matrices is invertible. ■

4.2 Deriving the algorithm

Start from equation (3), $E_k \cdots E_2 E_1 A = I_n$, and multiply both sides on the **right** by A^{-1}:

$$E_k \cdots E_2 E_1 \underbrace{A A^{-1}}_{I_n} = I_n A^{-1} \qquad \Longrightarrow \qquad \boxed{A^{-1} = E_k \cdots E_2 E_1 \, I_n}$$

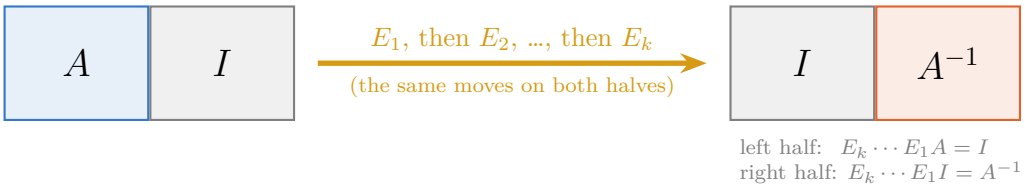
Read the boxed formula slowly: *the same sequence of row operations that reduces A to I_n will transform I_n into A^{-1}.*

Inversion algorithm [Anton §1.5]

To find the inverse of an invertible matrix A: write the partitioned matrix $[\,A \mid I\,]$ and apply row operations until the left side becomes I. The right side is then A^{-1}:

$$[\,A \mid I\,] \longrightarrow [\,I \mid A^{-1}\,].$$

If at some stage a **row of zeros** appears on the left side, stop: A is **not invertible**.



A good order of moves (Gauss–Jordan)

1. Make the $(1,1)$ entry a leading 1 (swap or scale if necessary), then make every entry **below** it zero.
2. Move to the $(2,2)$ entry and do the same, then $(3,3)$, … (now the left side is upper triangular).
3. Working from the bottom up, make every entry **above** each leading 1 zero.

4.3 Example 4: the inverse *and* its elementary matrices

Example 4 Using row operations to find A^{-1} [Anton §1.5 · Ex. 4]

Find the inverse of $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$.

Solution. We adjoin I_3 to the right of A and reduce. The book does two moves in some steps; here every **single** move is shown, so we can name its elementary matrix.

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array}\right] \quad \text{(start: } [A \,|\, I]\text{)}$$

$$\xrightarrow[E_1]{R_2 \to R_2 - 2R_1} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ \mathbf{0} & \mathbf{1} & \mathbf{-3} & \mathbf{-2} & \mathbf{1} & \mathbf{0} \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array}\right] \xrightarrow[E_2]{R_3 \to R_3 - R_1} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ \mathbf{0} & \mathbf{-2} & \mathbf{5} & \mathbf{-1} & \mathbf{0} & \mathbf{1} \end{array}\right]$$

$$\xrightarrow[E_3]{R_3 \to R_3 + 2R_2} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ \mathbf{0} & \mathbf{0} & \mathbf{-1} & \mathbf{-5} & \mathbf{2} & \mathbf{1} \end{array}\right] \xrightarrow[E_4]{R_3 \to -R_3} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{5} & \mathbf{-2} & \mathbf{-1} \end{array}\right]$$

$$\xrightarrow[E_5]{R_2 \to R_2 + 3R_3} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ \mathbf{0} & \mathbf{1} & \mathbf{0} & \mathbf{13} & \mathbf{-5} & \mathbf{-3} \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right] \xrightarrow[E_6]{R_1 \to R_1 - 3R_3} \left[\begin{array}{ccc|ccc} \mathbf{1} & \mathbf{2} & \mathbf{0} & \mathbf{-14} & \mathbf{6} & \mathbf{3} \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right]$$

$$\xrightarrow[E_7]{R_1 \to R_1 - 2R_2} \left[\begin{array}{ccc|ccc} \mathbf{1} & \mathbf{0} & \mathbf{0} & \mathbf{-40} & \mathbf{16} & \mathbf{9} \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right]$$

The left side is now I_3, so

$$A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}.$$

Check $AA^{-1} = I_3$ (each row of A times each column of A^{-1}):

row 1: $1(-40) + 2(13) + 3(5) = 1$, $1(16) + 2(-5) + 3(-2) = 0$, $1(9) + 2(-3) + 3(-1) = 0$,
row 2: $2(-40) + 5(13) + 3(5) = 0$, $2(16) + 5(-5) + 3(-2) = 1$, $2(9) + 5(-3) + 3(-1) = 0$,
row 3: $1(-40) + 0(13) + 8(5) = 0$, $1(16) + 0(-5) + 8(-2) = 0$, $1(9) + 0(-3) + 8(-1) = 1$.

The seven elementary matrices hidden in Example 4

Each arrow above is one elementary matrix: do the same move to I_3. Its inverse comes from the inverse move (Table 1).

Step	Move	Elementary matrix	Inverse move	Inverse matrix
1	$R_2 \to R_2 - 2R_1$	$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$R_2 \to R_2 + 2R_1$	$E_1^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
2	$R_3 \to R_3 - R_1$	$E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$	$R_3 \to R_3 + R_1$	$E_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$
3	$R_3 \to R_3 + 2R_2$	$E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix}$	$R_3 \to R_3 - 2R_2$	$E_3^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}$
4	$R_3 \to -R_3$	$E_4 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$	$R_3 \to -R_3$	$E_4^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$
5	$R_2 \to R_2 + 3R_3$	$E_5 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$	$R_2 \to R_2 - 3R_3$	$E_5^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix}$
6	$R_1 \to R_1 - 3R_3$	$E_6 = \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$R_1 \to R_1 + 3R_3$	$E_6^{-1} = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
7	$R_1 \to R_1 - 2R_2$	$E_7 = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$R_1 \to R_1 + 2R_2$	$E_7^{-1} = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

What the algorithm really computed

$$E_7E_6E_5E_4E_3E_2E_1\,A = I_3 \quad \Longrightarrow \quad \boxed{A^{-1} = E_7E_6E_5E_4E_3E_2E_1}$$

and (multiply by the inverses in reverse order) $\boxed{A = E_1^{-1}E_2^{-1}E_3^{-1}E_4^{-1}E_5^{-1}E_6^{-1}E_7^{-1}}$

Nice fact: after step k, the right half of the augmented matrix is exactly the product $E_k \cdots E_2E_1$. After step 1 the right half is E_1, and after step 2 it is

$$E_2E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

— exactly what we saw. So the right half is a *running record* of all moves made so far.

Common mistake

- **Order!** The first move E_1 is written on the **right** in $A^{-1} = E_7 \cdots E_1$, but on the **left** in $A = E_1^{-1} \cdots E_7^{-1}$.
- The factorisation is **not unique**: a different sequence of moves gives different elementary matrices [Anton TF (g)].

4.4 A complete 2×2 example: A and A^{-1} as products

Example **Express A and A^{-1} as products of elementary matrices** [Anton §1.5 · Ex. Set 1.5, No. 23]

$A = \begin{bmatrix} -3 & 1 \\ 2 & 2 \end{bmatrix}$.

Step 1: reduce $[\,A \mid I\,]$ and record every move.

$$\left[\begin{array}{cc|cc} -3 & 1 & 1 & 0 \\ 2 & 2 & 0 & 1 \end{array}\right] \xrightarrow[E_1]{R_1 \leftrightarrow R_2} \left[\begin{array}{cc|cc} \mathbf{2} & \mathbf{2} & \mathbf{0} & \mathbf{1} \\ \mathbf{-3} & \mathbf{1} & \mathbf{1} & \mathbf{0} \end{array}\right] \xrightarrow[E_2]{R_1 \to \frac{1}{2}R_1} \left[\begin{array}{cc|cc} \mathbf{1} & \mathbf{1} & \mathbf{0} & \mathbf{\frac{1}{2}} \\ -3 & 1 & 1 & 0 \end{array}\right]$$

$$\xrightarrow[E_3]{R_2 \to R_2 + 3R_1} \left[\begin{array}{cc|cc} 1 & 1 & 0 & \frac{1}{2} \\ \mathbf{0} & \mathbf{4} & \mathbf{1} & \mathbf{\frac{3}{2}} \end{array}\right] \xrightarrow[E_4]{R_2 \to \frac{1}{4}R_2} \left[\begin{array}{cc|cc} 1 & 1 & 0 & \frac{1}{2} \\ \mathbf{0} & \mathbf{1} & \mathbf{\frac{1}{4}} & \mathbf{\frac{3}{8}} \end{array}\right]$$

$$\xrightarrow[E_5]{R_1 \to R_1 - R_2} \left[\begin{array}{cc|cc} \mathbf{1} & \mathbf{0} & \mathbf{-\frac{1}{4}} & \mathbf{\frac{1}{8}} \\ 0 & 1 & \frac{1}{4} & \frac{3}{8} \end{array}\right]$$

So

$$A^{-1} = \begin{bmatrix} -\frac{1}{4} & \frac{1}{8} \\ \frac{1}{4} & \frac{3}{8} \end{bmatrix}.$$

Check with the 2×2 formula: $\det A = (-3)(2) - (1)(2) = -8$ and

$$A^{-1} = \frac{1}{-8}\begin{bmatrix} 2 & -1 \\ -2 & -3 \end{bmatrix} = \begin{bmatrix} -\frac{1}{4} & \frac{1}{8} \\ \frac{1}{4} & \frac{3}{8} \end{bmatrix}. \checkmark$$

Step 2: the elementary matrices (same move on I_2) and their inverses.

$$E_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \ (R_1 \leftrightarrow R_2) \qquad E_2 = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{bmatrix} \ (R_1 \to \tfrac{1}{2}R_1) \qquad E_3 = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \ (R_2 \to R_2 + 3R_1)$$

$$E_4 = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{4} \end{bmatrix} \ (R_2 \to \tfrac{1}{4}R_2) \qquad E_5 = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \ (R_1 \to R_1 - R_2)$$

$$E_1^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \qquad E_2^{-1} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \qquad E_3^{-1} = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$$

$$E_4^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix} \qquad E_5^{-1} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Step 3: $A^{-1} = E_5E_4E_3E_2E_1$ (multiply from right to left; every product is shown).

$$E_2E_1 = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{2} \\ 1 & 0 \end{bmatrix}$$

$$E_3(E_2E_1) = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}\begin{bmatrix} 0 & \frac{1}{2} \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{2} \\ 1 & \frac{3}{2} \end{bmatrix}$$

$$E_4(E_3E_2E_1) = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{4} \end{bmatrix}\begin{bmatrix} 0 & \frac{1}{2} \\ 1 & \frac{3}{2} \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{2} \\ \frac{1}{4} & \frac{3}{8} \end{bmatrix}$$

$$E_5(E_4E_3E_2E_1) = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 0 & \frac{1}{2} \\ \frac{1}{4} & \frac{3}{8} \end{bmatrix} = \begin{bmatrix} -\frac{1}{4} & \frac{1}{8} \\ \frac{1}{4} & \frac{3}{8} \end{bmatrix}$$

which is A^{-1}. ✓
Step 4: $A = E_1^{-1}E_2^{-1}E_3^{-1}E_4^{-1}E_5^{-1}$ (now build from the right end).

$$E_4^{-1}E_5^{-1} = \begin{bmatrix}1 & 0\\ 0 & 4\end{bmatrix}\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 1\\ 0 & 4\end{bmatrix}$$

$$E_3^{-1}(E_4^{-1}E_5^{-1}) = \begin{bmatrix}1 & 0\\ -3 & 1\end{bmatrix}\begin{bmatrix}1 & 1\\ 0 & 4\end{bmatrix} = \begin{bmatrix}1 & 1\\ -3 & 1\end{bmatrix}$$

$$E_2^{-1}(E_3^{-1}E_4^{-1}E_5^{-1}) = \begin{bmatrix}2 & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & 1\\ -3 & 1\end{bmatrix} = \begin{bmatrix}2 & 2\\ -3 & 1\end{bmatrix}$$

$$E_1^{-1}(E_2^{-1}E_3^{-1}E_4^{-1}E_5^{-1}) = \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}\begin{bmatrix}2 & 2\\ -3 & 1\end{bmatrix} = \begin{bmatrix}-3 & 1\\ 2 & 2\end{bmatrix}$$

which is A. ✓ So

$$\boxed{\begin{bmatrix}-3 & 1\\ 2 & 2\end{bmatrix} = \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}\begin{bmatrix}2 & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ -3 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & 4\end{bmatrix}\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}}$$

4.5 When there is no inverse

Example 5 Showing that a matrix is not invertible [Anton §1.5 · Ex. 5]

$A = \begin{bmatrix}1 & 6 & 4\\ 2 & 4 & -1\\ -1 & 2 & 5\end{bmatrix}$. Apply the procedure of Example 4:

$$\left[\begin{array}{ccc|ccc}1 & 6 & 4 & 1 & 0 & 0\\ 2 & 4 & -1 & 0 & 1 & 0\\ -1 & 2 & 5 & 0 & 0 & 1\end{array}\right] \xrightarrow[E_1]{R_2\to R_2-2R_1} \left[\begin{array}{ccc|ccc}1 & 6 & 4 & 1 & 0 & 0\\ \mathbf{0} & \mathbf{-8} & \mathbf{-9} & \mathbf{-2} & \mathbf{1} & \mathbf{0}\\ -1 & 2 & 5 & 0 & 0 & 1\end{array}\right]$$

$$\xrightarrow[E_2]{R_3\to R_3+R_1} \left[\begin{array}{ccc|ccc}1 & 6 & 4 & 1 & 0 & 0\\ 0 & -8 & -9 & -2 & 1 & 0\\ \mathbf{0} & \mathbf{8} & \mathbf{9} & \mathbf{1} & \mathbf{0} & \mathbf{1}\end{array}\right] \xrightarrow[E_3]{R_3\to R_3+R_2} \left[\begin{array}{ccc|ccc}1 & 6 & 4 & 1 & 0 & 0\\ 0 & -8 & -9 & -2 & 1 & 0\\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{-1} & \mathbf{1} & \mathbf{1}\end{array}\right]$$

Since we have obtained a **row of zeros on the left side**, A is **not invertible**. (By Theorem 1.5.3: $\text{rref}(A) \neq I_3$, so (c) is false, hence (a) is false.)

Example 6 Analysing homogeneous systems [Anton §1.5 · Ex. 6]

Do these systems have nontrivial solutions?

(a)
$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 0\\ 2x_1 + 5x_2 + 3x_3 &= 0\\ x_1 \qquad\quad + 8x_3 &= 0\end{aligned}$$

(b)
$$\begin{aligned} x_1 + 6x_2 + 4x_3 &= 0\\ 2x_1 + 4x_2 - x_3 &= 0\\ -x_1 + 2x_2 + 5x_3 &= 0\end{aligned}$$

Solution. By (a)⇔(b) of Theorem 1.5.3, $A\mathbf{x} = \mathbf{0}$ has only the trivial solution exactly when A is invertible. The coefficient matrix of (a) is the matrix of Example 4 (invertible), so (a) has **only the trivial solution**. The coefficient matrix of (b) is the matrix of Example 5 (not invertible), so (b) has **nontrivial solutions**.

5 Elementary matrices *without* finding an inverse

5.1 From a single row operation, directly

You never need an inverse to write an elementary matrix: **do the move to** I. If A is $m \times n$, use I_m.

Where does the number go?

- $R_i \leftrightarrow R_j$: swap rows i and j of I (the 1s move to positions (i,j) and (j,i)).
- $R_i \to cR_i$: put c in the diagonal position (i,i).
- $R_i \to R_i + cR_j$: put c in position (i,j) — **row** = the row that changes, **column** = the row you borrow from.

Example Writing E straight from the move

For a matrix with 4 rows:

$$R_2 \to R_2 - 4R_3: \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & \mathbf{-4} & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad R_1 \leftrightarrow R_4: \begin{bmatrix} \mathbf{0} & 0 & 0 & \mathbf{1} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \mathbf{1} & 0 & 0 & \mathbf{0} \end{bmatrix} \quad R_3 \to \tfrac{1}{2}R_3: \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \mathbf{\frac{1}{2}} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

For $R_2 \to R_2 - 4R_3$ the -4 sits in row 2 (the row that changes), column 3 (the row we borrow from).

Example Identify the move and verify EA [Anton §1.5 · Ex. Set 1.5, No. 5(b)]

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix}, \quad A = \begin{bmatrix} 2 & -1 & 0 & -4 & -4 \\ 1 & -3 & -1 & 5 & 3 \\ 2 & 0 & 1 & 3 & -1 \end{bmatrix}.$$

Solution. E differs from I_3 only in position $(3,2)$, which is -3. So E comes from $R_3 \to R_3 - 3R_2$. Rows 1, 2 of EA are copies of rows 1, 2 of A; row 3 is

$$\begin{aligned} R_3 - 3R_2 &= (2,0,1,3,-1) - 3(1,-3,-1,5,3) \\ &= (2-3,\ 0+9,\ 1+3,\ 3-15,\ -1-9) = (-1,9,4,-12,-10), \end{aligned}$$

$$EA = \begin{bmatrix} 2 & -1 & 0 & -4 & -4 \\ 1 & -3 & -1 & 5 & 3 \\ \mathbf{-1} & \mathbf{9} & \mathbf{4} & \mathbf{-12} & \mathbf{-10} \end{bmatrix},$$

which is exactly A after the move $R_3 \to R_3 - 3R_2$. ✓

5.2 Finding E with $EA = B$ by comparing rows

Example **Find an elementary matrix E** [Anton §1.5 · Ex. Set 1.5, No. 7]

$$A = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 8 & 1 & 5 \end{bmatrix}, \qquad B = \begin{bmatrix} 8 & 1 & 5 \\ 2 & -7 & -1 \\ 3 & 4 & 1 \end{bmatrix}, \qquad C = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 2 & -7 & 3 \end{bmatrix}.$$

(a) $EA = B$. Row 2 is the same; row 1 of B is row 3 of A and row 3 of B is row 1 of A. So the move is $R_1 \leftrightarrow R_3$ and

$$E = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

Check:

$$EA = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 8 & 1 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 0\cdot 3 + 0\cdot 2 + 1\cdot 8 & 0\cdot 4 + 0(-7) + 1\cdot 1 & 0\cdot 1 + 0(-1) + 1\cdot 5 \\ 0\cdot 3 + 1\cdot 2 + 0\cdot 8 & 0\cdot 4 + 1(-7) + 0\cdot 1 & 0\cdot 1 + 1(-1) + 0\cdot 5 \\ 1\cdot 3 + 0\cdot 2 + 0\cdot 8 & 1\cdot 4 + 0(-7) + 0\cdot 1 & 1\cdot 1 + 0(-1) + 0\cdot 5 \end{bmatrix} = \begin{bmatrix} 8 & 1 & 5 \\ 2 & -7 & -1 \\ 3 & 4 & 1 \end{bmatrix} = B. \checkmark$$

(b) $EB = A$. Swapping rows 1 and 3 again undoes it, so the *same* E works (a swap is its own inverse).

(c) $EA = C$. Rows 1 and 2 are unchanged; only row 3 changed. Find the change:

$$(\text{new row 3}) - (\text{old row 3}) = (2, -7, 3) - (8, 1, 5) = (-6, -8, -2) = -2\,(3, 4, 1) = -2R_1.$$

So the move is $R_3 \to R_3 - 2R_1$ and

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}.$$

Check: rows 1, 2 of EA are copied, and row 3 of EA is

$$(-2)(3, 4, 1) + 0(2, -7, -1) + 1(8, 1, 5) = (-6 + 8,\ -8 + 1,\ -2 + 5) = (2, -7, 3). \checkmark$$

(d) $EC = A$. Undo (c): change the sign. $R_3 \to R_3 + 2R_1$, $\quad E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$.

In simple words

Compare the two matrices row by row. Two rows exchanged ⇒ swap. One row multiplied by a number ⇒ scale. One row changed by a multiple of another row ⇒ subtract "new − old" and recognise it as $c\times$(some other row). (Another road is $E = BA^{-1}$, but that needs A^{-1} — comparing rows is much faster.)

5.3 Writing E for given operations

Example **Three operations on a 3×3 matrix** [Williams §2.4 · Ex. Set 2.4, No. 31]

Let $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$. T_1: interchange rows 1 and 2; T_2: multiply row 3 by -2; T_3: add 4 times row 1 to row 3. Show that they are performed by

$$E_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}, \qquad E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix}.$$

Solution. Each E_k is I_3 after T_k: $I_3 \xrightarrow{R_1 \leftrightarrow R_2} E_1$, $I_3 \xrightarrow{R_3 \to -2R_3} E_2$, $I_3 \xrightarrow{R_3 \to R_3 + 4R_1} E_3$. Multiplying (row of E = recipe):

$$E_1 A = \begin{bmatrix} \mathbf{d} & \mathbf{e} & \mathbf{f} \\ \mathbf{a} & \mathbf{b} & \mathbf{c} \\ g & h & i \end{bmatrix}, \qquad E_2 A = \begin{bmatrix} a & b & c \\ d & e & f \\ \mathbf{-2g} & \mathbf{-2h} & \mathbf{-2i} \end{bmatrix}, \qquad E_3 A = \begin{bmatrix} a & b & c \\ d & e & f \\ \mathbf{g + 4a} & \mathbf{h + 4b} & \mathbf{i + 4c} \end{bmatrix}.$$

For instance, row 3 of E_3 is $(4, 0, 1)$, so row 3 of $E_3 A$ is $4(a, b, c) + 0(d, e, f) + 1(g, h, i) = (g + 4a,\ h + 4b,\ i + 4c)$. ✓

Example **Elementary 3×3 matrices of two operations** [Williams §2.4 · Ex. Set 2.4, No. 32]

T_1: multiply row 1 by -3. T_2: add 3 times row 2 to row 1.

$$I_3 \xrightarrow{R_1 \to -3R_1} \begin{bmatrix} \mathbf{-3} & \mathbf{0} & \mathbf{0} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = E_1, \qquad I_3 \xrightarrow{R_1 \to R_1 + 3R_2} \begin{bmatrix} \mathbf{1} & \mathbf{3} & \mathbf{0} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = E_2.$$

Effect on $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$: $E_1 A = \begin{bmatrix} -3a & -3b & -3c \\ d & e & f \\ g & h & i \end{bmatrix}$, $E_2 A = \begin{bmatrix} a + 3d & b + 3e & c + 3f \\ d & e & f \\ g & h & i \end{bmatrix}$.

Inverses (Section 3): $E_1^{-1} = \begin{bmatrix} -\frac{1}{3} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $E_2^{-1} = \begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

5.4 Elimination without an inverse: the LU decomposition

Elementary matrices are also useful when A is only *partly* reduced. Stop Gaussian elimination when the matrix is **upper triangular** (zeros below the diagonal); call it U.

Example $A = LU$

$A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix}$. Make zeros below the diagonal, one move at a time:

$$\begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix} \xrightarrow[E_1]{R_2 \to R_2 - 2R_1} \begin{bmatrix} 2 & 1 & 1 \\ \mathbf{0} & \mathbf{-8} & \mathbf{-2} \\ -2 & 7 & 2 \end{bmatrix} \xrightarrow[E_2]{R_3 \to R_3 + R_1} \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ \mathbf{0} & \mathbf{8} & \mathbf{3} \end{bmatrix}$$

$$\xrightarrow[E_3]{R_3 \to R_3 + R_2} \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ \mathbf{0} & \mathbf{0} & \mathbf{1} \end{bmatrix}$$

The three elementary matrices (same move on I_3) and their inverses:

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}, \quad E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix};$$

$$E_1^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}, \quad E_3^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}.$$

So $E_3E_2E_1A = U$, and multiplying by the inverses (in reverse order),

$$A = \underbrace{E_1^{-1}E_2^{-1}E_3^{-1}}_{L}\, U.$$

Compute L (right to left):

$$E_2^{-1}E_3^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix},$$

$$L = E_1^{-1}\big(E_2^{-1}E_3^{-1}\big) = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix}.$$

(Row 3 of the first product: $(-1)(1,0,0) + 0 + 1(0,-1,1) = (-1,-1,1)$. In the second product row 2 is $2(1,0,0) + 1(0,1,0) = (2,1,0)$.)

Check $LU = A$ row by row (row of L = recipe for the rows of U):

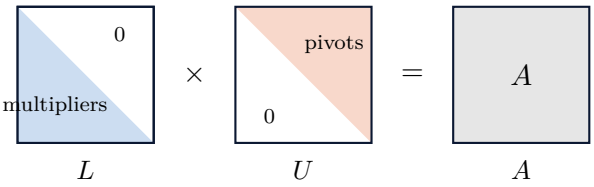
row 1: $1(2,1,1) = (2,1,1)$,

row 2: $2(2,1,1) + 1(0,-8,-2) = (4,2,2) + (0,-8,-2) = (4,-6,0)$,

row 3: $-1(2,1,1) - 1(0,-8,-2) + 1(0,0,1) = (-2,-1,-1) + (0,8,2) + (0,0,1) = (-2,7,2)$.

$$\begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix}}_{L}\underbrace{\begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{bmatrix}}_{U} \checkmark$$

L = lower triangular
U = upper triangular



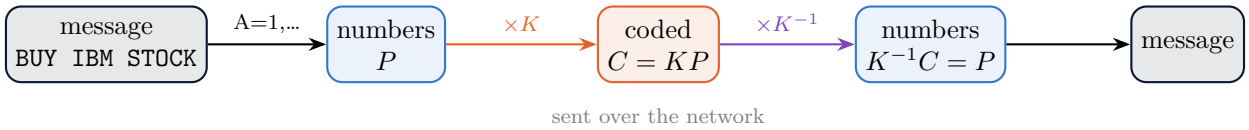
The entries of L below the diagonal are the multipliers **with the sign changed**:
$R_2 - 2R_1$ puts **2** in L;
$R_3 + R_1$ puts $-\mathbf{1}$; $R_3 + R_2$ puts $-\mathbf{1}$.

6 Applications of the inverse

6.1 Cryptography: hiding a message with an invertible matrix

[Williams §2.4]

Cryptography (Greek *kryptos*, "hidden") is the art of coding and decoding messages. One method uses an invertible matrix K (the **encoding matrix**); the receiver decodes with K^{-1} (the **decoding matrix**).



Step 1: letters to numbers. Each letter gets its position in the alphabet; a space is 27.

A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	2	3	4	5	6	7	8	9	10	11	12	13	14

O	P	Q	R	S	T	U	V	W	X	Y	Z	␣
15	16	17	18	19	20	21	22	23	24	25	26	27

B	U	Y		I	B	M		S	T	O	C	K		
2	21	25	27	9	2	13	27	19	20	15	3	11	27	27

Two extra spaces were added at the end so that the numbers fill complete columns of length 3.

Step 2: put the numbers in 3×1 columns, and the columns side by side:

$$\begin{bmatrix}2\\21\\25\end{bmatrix}, \begin{bmatrix}27\\9\\2\end{bmatrix}, \begin{bmatrix}13\\27\\19\end{bmatrix}, \begin{bmatrix}20\\15\\3\end{bmatrix}, \begin{bmatrix}11\\27\\27\end{bmatrix} \quad \Longrightarrow \quad P = \begin{bmatrix}2 & 27 & 13 & 20 & 11\\21 & 9 & 27 & 15 & 27\\25 & 2 & 19 & 3 & 27\end{bmatrix}.$$

Step 3: encode with the matrix

$$K = \begin{bmatrix}-3 & -3 & -4\\0 & 1 & 1\\4 & 3 & 4\end{bmatrix}.$$

(By cofactors along row 1: $\det K = -3(1\cdot4-1\cdot3)-(-3)(0\cdot4-1\cdot4)+(-4)(0\cdot3-1\cdot4) = -3-12+16 = 1$, so K is invertible.)

First column in full (row of K times column $(2, 21, 25)$):

$$\begin{aligned}(-3)(2)+(-3)(21)+(-4)(25) &= -6-63-100 = -169,\\ (0)(2)+(1)(21)+(1)(25) &= 0+21+25 = 46,\\ (4)(2)+(3)(21)+(4)(25) &= 8+63+100 = 171.\end{aligned}$$

Doing the same for every column:

$$C = KP = \begin{bmatrix} -3 & -3 & -4 \\ 0 & 1 & 1 \\ 4 & 3 & 4 \end{bmatrix} \begin{bmatrix} 2 & 27 & 13 & 20 & 11 \\ 21 & 9 & 27 & 15 & 27 \\ 25 & 2 & 19 & 3 & 27 \end{bmatrix} = \begin{bmatrix} -169 & -116 & -196 & -117 & -222 \\ 46 & 11 & 46 & 18 & 54 \\ 171 & 143 & 209 & 137 & 233 \end{bmatrix}.$$

The message is sent column by column:

$$-169,\ 46,\ 171,\ -116,\ 11,\ 143,\ -196,\ 46,\ 209,\ -117,\ 18,\ 137,\ -222,\ 54,\ 233.$$

To anyone without K^{-1} these numbers mean nothing. (Notice that the letter B, which appears twice, is coded differently each time.)

Step 4: find the decoding matrix K^{-1} with the inversion algorithm of Section 4:

$$\left[\begin{array}{ccc|ccc} -3 & -3 & -4 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 4 & 3 & 4 & 0 & 0 & 1 \end{array}\right] \quad \text{(start: } [K \mid I]\text{)}$$

$$\xrightarrow[E_1]{R_1 \to R_1 + R_3} \left[\begin{array}{ccc|ccc} \mathbf{1} & \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{0} & \mathbf{1} \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 4 & 3 & 4 & 0 & 0 & 1 \end{array}\right] \xrightarrow[E_2]{R_3 \to R_3 - 4R_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ \mathbf{0} & \mathbf{3} & \mathbf{4} & \mathbf{-4} & \mathbf{0} & \mathbf{-3} \end{array}\right]$$

$$\xrightarrow[E_3]{R_3 \to R_3 - 3R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{-4} & \mathbf{-3} & \mathbf{-3} \end{array}\right] \xrightarrow[E_4]{R_2 \to R_2 - R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 1 \\ \mathbf{0} & \mathbf{1} & \mathbf{0} & \mathbf{4} & \mathbf{4} & \mathbf{3} \\ 0 & 0 & 1 & -4 & -3 & -3 \end{array}\right]$$

$$K^{-1} = \begin{bmatrix} 1 & 0 & 1 \\ 4 & 4 & 3 \\ -4 & -3 & -3 \end{bmatrix}.$$

Check row 2 of KK^{-1}: $(0,1,1)$ times the columns: $0+4-4=0$, $0+4-3=1$, $0+3-3=0$. ✓

Step 5: decode. First received column in full:

$$\begin{aligned} (1)(-169) + (0)(46) + (1)(171) &= -169 + 0 + 171 = 2, \\ (4)(-169) + (4)(46) + (3)(171) &= -676 + 184 + 513 = 21, \\ (-4)(-169) + (-3)(46) + (-3)(171) &= 676 - 138 - 513 = 25. \end{aligned}$$

All columns:

$$K^{-1}C = \begin{bmatrix} 1 & 0 & 1 \\ 4 & 4 & 3 \\ -4 & -3 & -3 \end{bmatrix} \begin{bmatrix} -169 & -116 & -196 & -117 & -222 \\ 46 & 11 & 46 & 18 & 54 \\ 171 & 143 & 209 & 137 & 233 \end{bmatrix} = \begin{bmatrix} 2 & 27 & 13 & 20 & 11 \\ 21 & 9 & 27 & 15 & 27 \\ 25 & 2 & 19 & 3 & 27 \end{bmatrix} = P,$$

2	21	25	27	9	2	13	27	19	20	15	3	11	27	27
B	U	Y		I	B	M		S	T	O	C	K		

In simple words

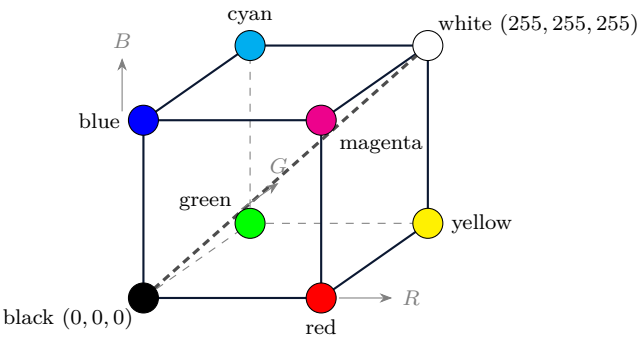
Why choose $\det K = 1$? Then K^{-1} also has **integer** entries, so the decoded numbers are whole numbers with no rounding. Bigger matrices make the code much harder to break.

6.2 Colour models: RGB ↔ YIQ

[Williams §2.4]

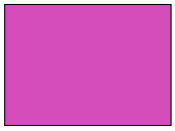
A computer monitor mixes **R**ed, **G**reen and **B**lue light; each value is a whole number from 0 to 255 (one byte, $2^8 = 256$). Analogue (NTSC) television used instead the **YIQ** model: Y = brightness (luma), I and Q = colour information. The NTSC encoder is a matrix transformation:

$$\begin{bmatrix} Y \\ I \\ Q \end{bmatrix} = \underbrace{\begin{bmatrix} .299 & .587 & .114 \\ .596 & -.275 & -.321 \\ .212 & -.523 & .311 \end{bmatrix}}_{T} \begin{bmatrix} R \\ G \\ B \end{bmatrix}.$$



the RGB cube $[0, 255]^3$

dashed diagonal: the greys $R = G = B$



purple text colour
RGB (213, 77, 187)
YIQ (130.204, 45.746, 63.042)



black: RGB = YIQ = $(0, 0, 0)$



black-and-white TV: YIQ $(s, 0, 0) \mapsto$ RGB (s, s, s)

Example Purple: from the computer to the television

Microsoft Word's purple has $R = 213$, $G = 77$, $B = 187$. Row by row:

$$\begin{aligned} Y &= .299(213) + .587(77) + .114(187) = 63.687 + 45.199 + 21.318 = 130.204, \\ I &= .596(213) - .275(77) - .321(187) = 126.948 - 21.175 - 60.027 = 45.746, \\ Q &= .212(213) - .523(77) + .311(187) = 45.156 - 40.271 + 58.157 = 63.042. \end{aligned}$$

These are the YIQ values used to reproduce this purple on a television screen. For black, $R = G = B = 0$ gives $Y = I = Q = 0$: black has the same RGB and YIQ values.

Back from TV to the computer: use T^{-1}. A YIQ signal is converted to RGB with the inverse matrix (to three decimal places):

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = T^{-1} \begin{bmatrix} Y \\ I \\ Q \end{bmatrix}, \qquad T^{-1} = \begin{bmatrix} .299 & .587 & .114 \\ .596 & -.275 & -.321 \\ .212 & -.523 & .311 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & .956 & .620 \\ 1 & -.272 & -.647 \\ 1 & -1.108 & 1.705 \end{bmatrix}.$$

Check with purple: $R = 130.204 + .956(45.746) + .620(63.042) = 130.204 + 43.733 + 39.086 = 213.02 \approx 213$, $G = 130.204 - .272(45.746) - .647(63.042) = 130.204 - 12.443 - 40.788 = 76.97 \approx 77$, $B = 130.204 - 1.108(45.746) + 1.705(63.042) = 130.204 - 50.687 + 107.487 = 187.00 \approx 187$. ✓ (Small differences come only from rounding T^{-1} to three decimals; with the exact inverse we get $213, 77, 187$ exactly.)

Example Black-and-white television [Williams §2.4 · Ex. Set 2.4, No. 36]

A black-and-white set uses only Y. A signal $(s, 0, 0)$ becomes

$$T^{-1} \begin{bmatrix} s \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & .956 & .620 \\ 1 & -.272 & -.647 \\ 1 & -1.108 & 1.705 \end{bmatrix} \begin{bmatrix} s \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \cdot s + 0 + 0 \\ 1 \cdot s + 0 + 0 \\ 1 \cdot s + 0 + 0 \end{bmatrix} = \begin{bmatrix} s \\ s \\ s \end{bmatrix},$$

a grey. This happens because the **first column of T^{-1} is all ones**. So $(255,0,0)$, $(200,0,0)$, $(150,0,0)$, $(100,0,0)$, $(0,0,0)$ give RGB $(255,255,255)$ (white), $(200,200,200)$, $(150,150,150)$, $(100,100,100)$, $(0,0,0)$ (black): decreasing s from 255 to 0 runs from white through greys to black (see the swatches above).

Range of Y, I, Q [Williams No. 34]

With $0 \le R,G,B \le 255$: the largest Y takes all three at 255, so $0 \le Y \le (.299+.587+.114)255 = 255$. For I the positive coefficient is .596 and the negative ones are $-.275, -.321$, so $-151.98 \le I \le 151.98$. For Q: $-.523(255) = -133.365 \le Q \le (.212+.311)255 = 133.365$.

7 Applications of elementary matrices

7.1 LU decomposition: how computers (and MATLAB) solve $A\mathbf{x} = \mathbf{b}$

Williams: *"this way of implementing row operations is particularly appropriate for computers."* Once $A = LU$ is known (Section 5.4), the system $A\mathbf{x} = \mathbf{b}$, i.e. $L(U\mathbf{x}) = \mathbf{b}$, is solved in two easy triangular steps:

$$\text{(1) solve } L\mathbf{y} = \mathbf{b} \text{ (forward substitution)}, \qquad \text{(2) solve } U\mathbf{x} = \mathbf{y} \text{ (back substitution)}.$$

Example Solve $A\mathbf{x} = \mathbf{b}$ with $A = LU$ from Section 5.4

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 5 \\ -2 \\ 9 \end{bmatrix}, \quad L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{bmatrix}.$$

(1) $L\mathbf{y} = \mathbf{b}$, top to bottom:

$$\begin{aligned} y_1 &= 5, \\ 2y_1 + y_2 = -2 \;&\Rightarrow\; y_2 = -2 - 2(5) = -12, \\ -y_1 - y_2 + y_3 = 9 \;&\Rightarrow\; y_3 = 9 + y_1 + y_2 = 9 + 5 - 12 = 2. \end{aligned}$$

(2) $U\mathbf{x} = \mathbf{y} = (5,-12,2)$, bottom to top:

$$\begin{aligned} x_3 &= 2, \\ -8x_2 - 2x_3 = -12 \;&\Rightarrow\; -8x_2 = -12 + 4 = -8 \;\Rightarrow\; x_2 = 1, \\ 2x_1 + x_2 + x_3 = 5 \;&\Rightarrow\; 2x_1 = 5 - 1 - 2 = 2 \;\Rightarrow\; x_1 = 1. \end{aligned}$$

So $\mathbf{x} = (1,1,2)$. Check: $A\mathbf{x} = (2+1+2,\ 4-6+0,\ -2+7+4) = (5,-2,9) = \mathbf{b}$. ✓

Where is this used?

- **MATLAB**, probably the most widely used matrix software, relies on LU decomposition: the command `[L,U,P] = lu(A)` returns L, U and a *permutation matrix* P (a product of swap matrices) with $PA = LU$. The backslash command `x = A\b` uses it too.
- The expensive part (finding L and U) is done **once**; each new right-hand side $\mathbf{b}$ then costs only two cheap substitutions. Engineering software solves the same A with thousands of different $\mathbf{b}$'s.
- LU also gives determinants quickly: $\det A = \det L \cdot \det U = 1 \cdot (2)(-8)(1) = -16$ here.

7.2 Theory: row equivalence and invertibility

Elementary matrices turn statements about row operations into algebra. If B is row equivalent to A, then

$$B = E_n \cdots E_2 E_1 A = QA, \qquad Q = E_n \cdots E_1 \text{ invertible.}$$

Theorem (Williams §2.4; Anton Ex. Set 1.5 No. 32)

If A and B are row equivalent matrices and A is invertible, then B is invertible.

Proof [Williams §2.4]. Since A and B are row equivalent, there are row operations $T_1, \dots, T_n$ with $B = T_n \circ \cdots \circ T_1(A)$. Let $E_1, \dots, E_n$ be their elementary matrices; then

$$B = E_n \cdots E_1 A.$$

The matrices $A, E_1, \dots, E_n$ are all invertible. Using $(XY)^{-1} = Y^{-1}X^{-1}$ again and again:

$$\begin{aligned} A^{-1}E_1^{-1}E_2^{-1}E_3^{-1}\cdots E_n^{-1} &= (E_1A)^{-1}E_2^{-1}E_3^{-1}\cdots E_n^{-1} \\ &= (E_2E_1A)^{-1}E_3^{-1}\cdots E_n^{-1} \\ &\;\;\vdots \\ &= (E_n\cdots E_1A)^{-1} = B^{-1}. \end{aligned}$$

Thus B is invertible and $\boxed{B^{-1} = A^{-1}E_1^{-1}E_2^{-1}\cdots E_n^{-1}}$. ∎

In simple words

Row operations can never destroy (or create) invertibility. That is why the inversion algorithm is allowed to stop as soon as a zero row appears: the current matrix is not invertible, so neither is A.

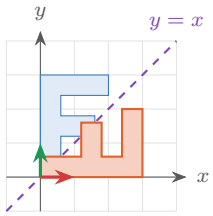
7.3 Graphics: every invertible map is a chain of simple moves

By Theorem 1.5.3, every invertible matrix is a product of elementary matrices. In the plane each elementary matrix is a *reflection*, a *stretch/compression* or a *shear* (Section 8). Therefore **every invertible linear map of the plane is a sequence of reflections, stretches and shears**. Graphics programs and fonts use exactly these building blocks ("skew", "flip", "scale").

8 Elementary matrices in 2D and 3D: the pictures

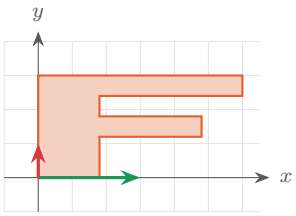
In every picture the **blue** F is the original and the **orange** F is its image under $\mathbf{x} \mapsto E\mathbf{x}$. The **green** and **red** arrows are $E\mathbf{e}_1$ and $E\mathbf{e}_2$ (the columns of E).

8.1 The 2×2 elementary matrices



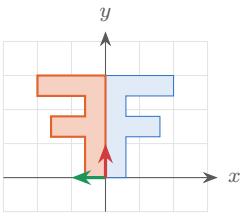
swap $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$: reflection in $y = x$

$\det = -1$ (area kept, flipped)



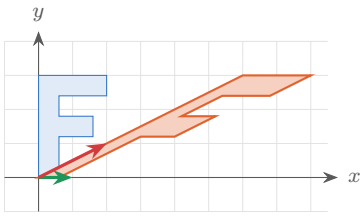
scale $\begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$: stretch $\times 3$ in x

$\det = 3$ (area $\times 3$)



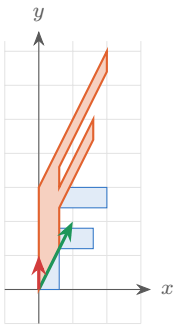
scale $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$: reflection in y-axis

$\det = -1$ (flipped)



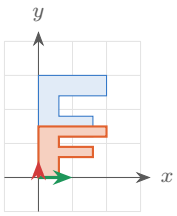
add $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ $(R_1 \to R_1 + 2R_2)$:

horizontal shear, $\det = 1$



add $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ $(R_2 \to R_2 + 2R_1)$:

vertical shear, $\det = 1$



scale $\begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$: compress in y

$\det = \frac{1}{2}$ (area halved)

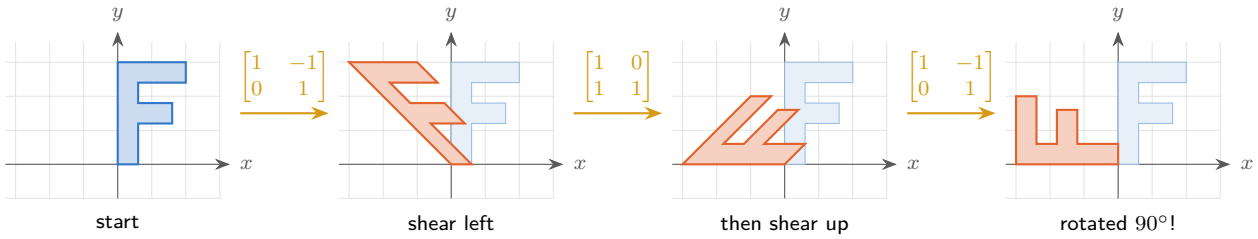
Dictionary: algebra $\leftrightarrow$ geometry in $\mathbb{R}^2$

Row operation	Matrix	Geometric effect
$R_1 \leftrightarrow R_2$	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	reflection in the line $y = x$ (swaps x and y)
$R_1 \to cR_1$	$\begin{bmatrix} c & 0 \\ 0 & 1 \end{bmatrix}$	stretch ($c > 1$) or compress ($0 < c < 1$) in the x-direction; also a reflection if $c < 0$
$R_2 \to cR_2$	$\begin{bmatrix} 1 & 0 \\ 0 & c \end{bmatrix}$	the same in the y-direction
$R_1 \to R_1 + kR_2$	$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$	shear in the x-direction: $(x, y) \mapsto (x + ky,\ y)$
$R_2 \to R_2 + kR_1$	$\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$	shear in the y-direction: $(x, y) \mapsto (x,\ y + kx)$

8.2 Three shears make a rotation

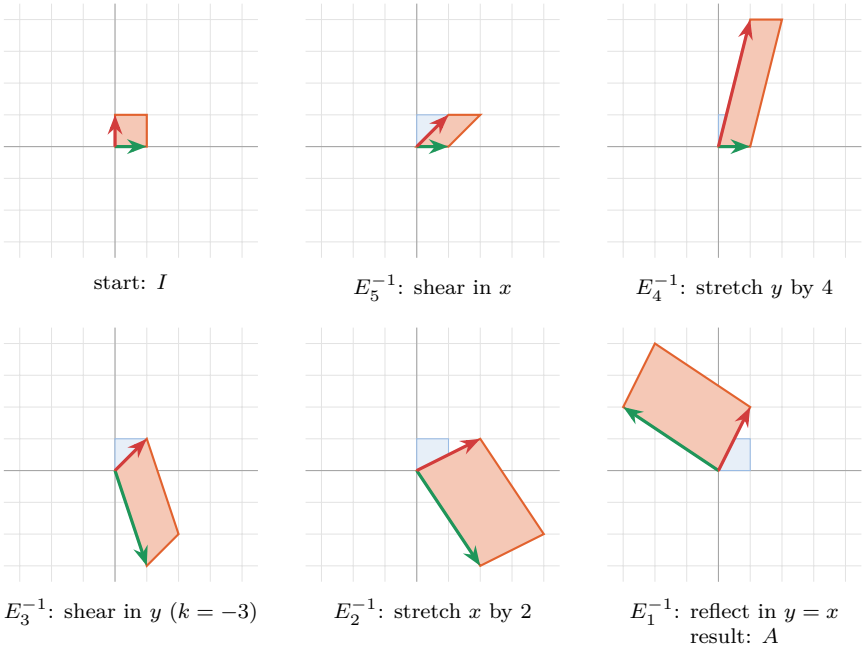
$$\begin{bmatrix}1 & -1\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}\begin{bmatrix}1 & -1\\ 0 & 1\end{bmatrix} = \begin{bmatrix}1 & -1\\ 0 & 1\end{bmatrix}\begin{bmatrix}1\cdot 1+0\cdot 0 & 1(-1)+0\cdot 1\\ 1\cdot 1+1\cdot 0 & 1(-1)+1\cdot 1\end{bmatrix} = \begin{bmatrix}1 & -1\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & -1\\ 1 & 0\end{bmatrix}$$
$$= \begin{bmatrix}1\cdot 1+(-1)\cdot 1 & 1(-1)+(-1)\cdot 0\\ 0\cdot 1+1\cdot 1 & 0(-1)+1\cdot 0\end{bmatrix} = \begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix},$$

which is the rotation by 90°. Read from right to left: first shear left, then shear up, then shear left again.



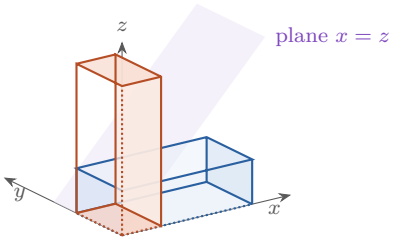
8.3 An invertible matrix as a chain of moves

From Section 4: $\begin{bmatrix}-3 & 1\\ 2 & 2\end{bmatrix} = E_1^{-1}E_2^{-1}E_3^{-1}E_4^{-1}E_5^{-1}$. Applied to a vector, the **rightmost** factor acts first. Watch the unit square:



The cumulative matrices are $\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}, \begin{bmatrix}1 & 1\\ 0 & 4\end{bmatrix}, \begin{bmatrix}1 & 1\\ -3 & 1\end{bmatrix}, \begin{bmatrix}2 & 2\\ -3 & 1\end{bmatrix}, \begin{bmatrix}-3 & 1\\ 2 & 2\end{bmatrix} = A$. The area of the final parallelogram is $|\det A| = |-6-2| = 8 = 1\cdot 4\cdot 1\cdot 2\cdot 1$ (only the two stretches change area).

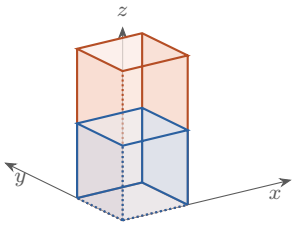
8.4 The 3×3 elementary matrices in space



swap $R_1 \leftrightarrow R_3$: $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

$(x, y, z) \mapsto (z, y, x)$: mirror in the plane $x = z$

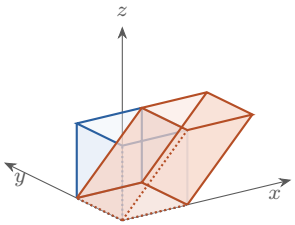
a $2 \times 1 \times 0.6$ brick becomes $0.6 \times 1 \times 2$; $\det = -1$



scale $R_3 \to 2R_3$: $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

$(x, y, z) \mapsto (x, y, 2z)$: stretch up

volume $\times 2$; $\det = 2$



add $R_1 \to R_1 + R_3$: $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$(x, y, z) \mapsto (x + z, y, z)$: shear

top slides by z; volume kept, $\det = 1$

(For the swap we used a brick instead of the cube, because the unit cube is symmetric in x and z and would look unchanged.)

8.5 Determinants of elementary matrices

Type	**Example**	$\det E$	**Meaning**
swap $R_i \leftrightarrow R_j$	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	-1	mirror: area/volume kept, orientation reversed
scale $R_i \to cR_i$	$\begin{bmatrix} c & 0 \\ 0 & 1 \end{bmatrix}$	c	area/volume multiplied by $\lvert c \rvert$
add $R_i \to R_i + kR_j$	$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$	1	shear: area/volume kept

$\det(EA) = \det(E)\det(A)$ — determinants from elimination

In Example 4 the seven moves had determinants $1, 1, 1, -1, 1, 1, 1$ (only E_4, $R_3 \to -R_3$, is not 1). From $E_7 \cdots E_1 A = I$:

$$1 \cdot 1 \cdot 1 \cdot (-1) \cdot 1 \cdot 1 \cdot 1 \cdot \det A = \det I = 1 \quad \Longrightarrow \quad \det A = -1.$$

Direct check (cofactors along row 1):

$$\det \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix} = 1(5 \cdot 8 - 3 \cdot 0) - 2(2 \cdot 8 - 3 \cdot 1) + 3(2 \cdot 0 - 5 \cdot 1) = 40 - 26 - 15 = -1. \checkmark$$

In No. 23: $\det E_1 \det E_2 \cdots \det E_5 = (-1)(\frac{1}{2})(1)(\frac{1}{4})(1) = -\frac{1}{8}$, so $\det A = -8$; indeed $(-3)(2) - (1)(2) = -8$. ✓

9 Everything on one page

Row operation	E **(on** I_3**)**	E^{-1}	$\det E$	**Geometry** ($\mathbb{R}^2/\mathbb{R}^3$)
swap $R_1 \leftrightarrow R_2$	$\begin{bmatrix}0&1&0\\1&0&0\\0&0&1\end{bmatrix}$	$\begin{bmatrix}0&1&0\\1&0&0\\0&0&1\end{bmatrix}$	-1	mirror ($E^{-1}=E$)
scale $R_2 \to cR_2$ $(c\neq 0)$	$\begin{bmatrix}1&0&0\\0&c&0\\0&0&1\end{bmatrix}$	$\begin{bmatrix}1&0&0\\0&\frac{1}{c}&0\\0&0&1\end{bmatrix}$	c	stretch / compress
add $R_3 \to R_3 + kR_1$	$\begin{bmatrix}1&0&0\\0&1&0\\k&0&1\end{bmatrix}$	$\begin{bmatrix}1&0&0\\0&1&0\\-k&0&1\end{bmatrix}$	1	shear

Build E**:** do ONE row operation to I_m (m = number of rows of A).
Use E**:** EA = that row operation on A (multiply on the **left**).

Invert E**:** inverse operation. Swap $\to$ same; scale $c \to 1/c$; add $k \to -k$. Every E is invertible.

Inverse of A**:** $[A \mid I] \to [I \mid A^{-1}]$. Zero row on the left $\Rightarrow$ no inverse.
$A^{-1} = E_k \cdots E_1, \quad A = E_1^{-1} \cdots E_k^{-1}$.

Equivalence: A invertible $\Leftrightarrow A\mathbf{x} = \mathbf{0}$ only trivially $\Leftrightarrow$ rref $A = I \Leftrightarrow A$ = product of elementary matrices.

LU**:** eliminate below the diagonal: $E_k \cdots E_1 A = U$; $L = E_1^{-1} \cdots E_k^{-1}$ (multipliers with sign changed).

Applications: codes $C = KP$, $P = K^{-1}C$; colour RGB$\leftrightarrow$YIQ with T, T^{-1}; graphics: reflect, stretch, shear.

Five sentences to remember

1. An elementary matrix is I after *one* row operation.
2. Multiplying by E on the left performs that row operation.
3. The inverse of E is the elementary matrix of the inverse operation.
4. The moves that turn A into I turn I into A^{-1}.
5. An invertible matrix is a product of elementary matrices, so geometrically it is a chain of reflections, stretches and shears.

10 Check yourself

Check yourself

1. Is $\begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}$ elementary? No: a zero row needs $c = 0$.
2. Write the 3×3 elementary matrix for $R_3 \to R_3 + 5R_1$. $\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 5 & 0 & 1\end{bmatrix}$
3. Write the inverse of $\begin{bmatrix}1 & 0\\ -7 & 1\end{bmatrix}$. $\begin{bmatrix}1 & 0\\ 7 & 1\end{bmatrix}$
4. Write the inverse of $\begin{bmatrix}1 & 0 & 0\\ 0 & -2 & 0\\ 0 & 0 & 1\end{bmatrix}$. $\begin{bmatrix}1 & 0 & 0\\ 0 & -\frac{1}{2} & 0\\ 0 & 0 & 1\end{bmatrix}$
5. Compute EA for $E = \begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix}$, $A = \begin{bmatrix}1 & 3\\ 0 & 1\end{bmatrix}$ without multiplying. $R_2 \to R_2 + 2R_1$: $\begin{bmatrix}1 & 3\\ 2 & 7\end{bmatrix}$
6. E swaps rows 1 and 3 of I_3. What is E^2? I_3 (swapping twice undoes it)
7. While reducing $[A \mid I]$ you get a zero row on the left. Conclusion? A is not invertible
8. In $A^{-1} = E_3E_2E_1$, which move was done first? E_1 (the rightmost)
9. Use the inversion algorithm on $\begin{bmatrix}1 & 4\\ 2 & 7\end{bmatrix}$. [Anton No. 9(a)] $R_2 - 2R_1$, $R_2 \to -R_2$, $R_1 - 4R_2$: $\begin{bmatrix}-7 & 4\\ 2 & -1\end{bmatrix}$
10. What does $\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}$ do to the plane? Its determinant? reflection in the x-axis; -1
11. Find E with $EA = B$ if B is A with row 2 multiplied by 4 (A is 3×5). $\begin{bmatrix}1 & 0 & 0\\ 0 & 4 & 0\\ 0 & 0 & 1\end{bmatrix}$
12. Decode the column $(-169, 46, 171)$ with K^{-1} of Section 6. $(2, 21, 25) =$ B U Y

Practice problems

Anton & Rorres, Exercise Set 1.5	
Nos. 1–2	Is the matrix elementary? (Section 1)
Nos. 3–4	Find a row operation and elementary matrix that restore I (Section 3)
Nos. 5–6	Identify the row operation of E and verify EA (Sections 2, 5)
Nos. 7–8	Find an elementary matrix E with $EA = B$, $EB = A$, … (Section 5)
Nos. 9–10	2×2 inverses: formula and inversion algorithm (Section 4)
Nos. 11–18	Inversion algorithm for 3×3 and 4×4 matrices (Section 4)
Nos. 19–22	Inverses with parameters k_i; values of c for which a matrix is invertible
Nos. 23–26	Express A and A^{-1} as products of elementary matrices (Section 4.4)
Nos. 27–28	Show A and B are row equivalent; find C with $CA = B$ (Sections 5, 7)
Williams, Exercise Set 2.4	
Nos. 31–33	Elementary matrices of given operations; quick rules for E^{-1} (Sections 3, 5)
Nos. 34–36	Colour models: range of Y, I, Q; YIQ → RGB; black-and-white TV (Section 6)
Nos. 37–42	Cryptography: encode, decode, find the encoding matrix (Section 6)